



تحوث – THOTH

Autonomous Interactive Robot for Smart Museum Guidance

Graduation Project Proposal 2026

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1 Project Overview

THOTH is an AI-powered Autonomous Mobile Robot (AMR) designed to serve as a multilingual, interactive guide for modern museums. The system is designed for deployment in large cultural institutions such as the Grand Egyptian Museum (GEM). Unlike static displays and traditional audio guides, THOTH can physically navigate museum environments, guiding visitors through exhibition spaces while providing explanations of artefacts.

The robot uses LiDAR and depth cameras to perceive its surroundings and navigate safely within complex indoor environments. Its navigation system is powered by the ROS2 Navigation Stack, enabling reliable localisation, mapping, and obstacle avoidance.

In addition to autonomous navigation, THOTH integrates Large Language Models (LLMs) and speech technologies to enable natural voice interaction with visitors. Through voice interaction and a built-in multimedia display, the robot can answer visitors' questions and present additional visual content, creating a more engaging museum experience.

2 Problem Statement

With the rapid growth of global tourism and increasing visitor numbers, museums face growing challenges in delivering engaging and accessible guided experiences. Museums today must support large and diverse audiences while maintaining meaningful and informative visitor interactions.

Passive Interaction: Traditional audio guides and mobile applications typically provide one-way information delivery, limiting visitor engagement and often going unnoticed during museum visits.

Human Constraints: Professional tour guides are limited in number and cannot efficiently support large visitor volumes or accommodate the wide range of languages required for international tourism.

Accessibility Barriers: Many traditional museum tours do not provide sufficient visual or real-time support for visitors with hearing impairments, reducing accessibility and inclusivity.

THOTH addresses these challenges by introducing an autonomous robotic guide capable of providing interactive, multilingual, and inclusive museum tours that can operate continuously and scale with visitor demand.

3 Unique Value of the Project

THOTH introduces a new approach to museum guidance by combining autonomous robotics with artificial intelligence to create an interactive and adaptive visitor experience.

Unlike traditional guidance systems that rely on static information delivery, THOTH actively guides visitors through exhibition spaces while responding to their questions in real time.

By integrating autonomous navigation with conversational AI, the system transforms museum tours from passive information consumption into interactive learning experiences. Visitors can engage directly with the robot and receive personalised explanations supported by multimedia content.

THOTH is also designed as a scalable platform that can be adapted for different museums, exhibitions, and visitor needs. New content, languages, and interaction features can be added without modifying the core robotic system, enabling long-term deployment in cultural and public spaces.

3.1 Comparison with Existing Solutions

Several solutions currently exist to assist museum visitors, including audio guides, mobile applications, information kiosks, and human tour guides. While these systems provide useful information, they differ significantly in interactivity, scalability, and the ability to guide visitors within exhibition spaces. Table 1 summarises the key differences between these approaches and the proposed THOTH robotic guide.

System	Interactive Dialogue	Physical Guidance	Scalability	Accessibility
Audio Guides	Limited	No	High	Low
Mobile Museum Apps	Limited	No	High	Moderate
Information Kiosks	Low	No	Moderate	Low
Human Tour Guides	High	Yes	Low	Moderate
THOTH Robot	High	Yes	High	High

Table 1: Comparison of THOTH with existing museum guidance solutions

3.2 THOTH Advantages

THOTH provides several advantages compared to traditional museum guidance systems:

- **Autonomous Guided Tours:** THOTH can navigate predefined routes within museum environments and guide visitors through exhibition areas while providing contextual explanations of artefacts.
- **Interactive Visitor Communication:** Visitors can ask questions and receive real-time responses through natural voice interaction, creating a more engaging and personalised experience.
- **Multimedia Explanations:** The system can present educational content using videos, images, and visual displays to enhance storytelling and visitor understanding.

- **Multilingual Communication:** THOTH can deliver information in multiple languages, making museum content accessible to international visitors.
- **Accessibility Support:** Real-time captions and visual explanations improve accessibility for visitors with hearing impairments.
- **Engaging Robotic Presence:** The physical presence of the robot creates a more memorable and interactive visitor experience compared to static information systems.

4 Technical Approach

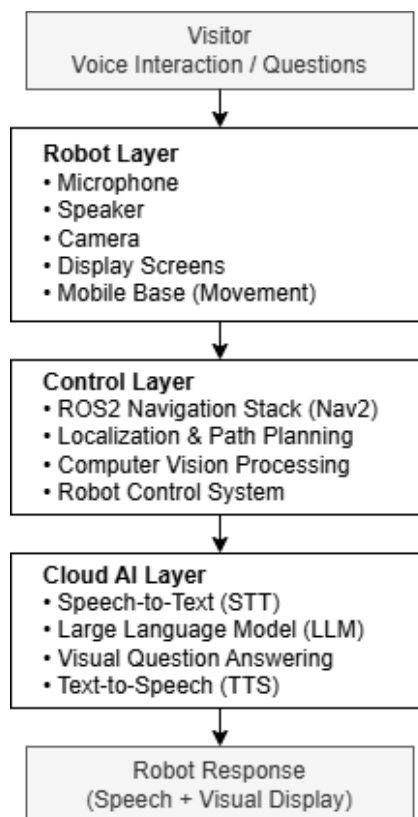


Figure 1: System architecture of the THOTH robotic tour guide

THOTH is designed as an integrated robotics platform that combines autonomous navigation, intelligent interaction, and multimedia communication to deliver an engaging museum-guidance experience.

To ensure reliable performance and modular system design, the robot follows a three-layer architecture consisting of the Robot Layer, Control Layer, and Cloud AI Layer. This structure separates hardware operation, real-time navigation, and high-level artificial intelligence processing, allowing each component to operate efficiently while remaining easily scalable.

4.1 Robot Layer (Physical Platform)

The Robot Layer represents the physical platform of THOTH and includes the hardware components responsible for sensing, mobility, and visitor interaction.

The robot is based on a differential-drive mobile base equipped with LiDAR and depth cameras to perceive its surroundings. These sensors enable the robot to detect obstacles, determine its position within the museum, and navigate safely through exhibition spaces.

To support human-robot interaction, the platform also includes a microphone array, high-fidelity speakers, and a multimedia display. These components enable the robot to communicate naturally with visitors and present visual information about artefacts and exhibitions.



(a) Conceptual design of THOTH



(b) Pudu Robotics reference service robot

Figure 2: Conceptual design and reference platform for the THOTH

4.2 Control Layer (Navigation and Perception)

The Control Layer is responsible for real-time robot operation, including localisation, mapping, and autonomous navigation. This layer is deployed on an NVIDIA Jetson Orin Nano running ROS2.

Using the ROS2 Navigation Stack (Nav2), the robot performs Simultaneous Localisation and Mapping (SLAM), path planning, and obstacle avoidance to move safely within dynamic museum environments. This layer ensures reliable navigation even in crowded exhibition halls with moving visitors.

4.3 Cloud AI Layer (Intelligent Interaction)

The Cloud AI Layer acts as the system’s cognitive engine, enabling natural, intelligent interaction between visitors and the robot.

This layer integrates Speech-to-Text (STT), Large Language Models (LLMs), and Text-to-Speech (TTS) technologies to support conversational dialogue with visitors. Through

voice interaction, visitors can ask questions about artefacts and receive real-time explanations.

Additionally, the system’s Visual Question Answering (VQA) capabilities enable it to interpret artifact-related visual queries and provide informative responses. By combining conversational AI with visual understanding, THOTH delivers a dynamic and engaging learning experience for museum visitors.

5 Implementation Timeline

The development of the THOTH robotic tour guide will take place over a six-month period from April to September. The project follows a structured development plan that progresses from system design and hardware integration to software development, system testing, and final demonstration. Each phase focuses on a specific component of the system to ensure systematic development and successful integration of all hardware and software modules.

Phase	Period	Main Activities
Research & System Design	April	Define system architecture, select hardware components, and plan software architecture.
Hardware Assembly	May	Build the robot structure, integrate sensors, computing unit, and mobility system.
Navigation Development	June – July	Implement SLAM, localization, path planning, and obstacle avoidance using ROS2 Nav2.
AI Interaction Development	August	Integrate speech recognition, conversational AI, and multimedia interaction systems.
Integration & Testing	September	Perform full system integration, conduct testing in simulated museum environments, and prepare the final demonstration.

Table 2: Proposed development timeline for the THOTH robotic tour guide

6 Detailed Budget Breakdown

The development of the THOTH robotic tour guide requires several hardware and system components that support autonomous navigation, visitor interaction, and multimedia presentation.

The following budget estimate outlines the primary subsystems required to build the robot prototype, including computing hardware, sensing equipment, mobility compo-

nents, and interaction interfaces. The listed costs represent approximate market prices and may be adjusted during the detailed design phase depending on component availability and testing results.

Category	Main Components	Estimated Cost (EGP)
AI Computing Unit	NVIDIA Jetson Orin Nano Developer Kit	35,000
Navigation Sensors	RPLiDAR A2/A3	8,000
Depth Camera	OAK-D Lite / Intel RealSense	8,000
Interaction System	Microphone array, speakers, multimedia display	5,000
Mobility System	Differential drive motors, wheels, encoders, motor drivers	5,000
Battery System	LiFePO4 battery pack, BMS, charging system	4,500
Mechanical Structure	Aluminum/acrylic frame and mounting components	4,000
Electronics & Wiring	Controllers, regulators, connectors, wiring	2,000
Miscellaneous Components	Mounts, adapters, safety parts	2,500

Table 3: Estimated hardware budget for the THOTH robotic platform

Based on the estimated component costs, the total hardware budget for the THOTH robotic prototype is approximately:

70,000 – 80,000 EGP

To account for price variations, shipping costs, and additional minor components, a contingency allocation of approximately 10–12% is included in the final funding request.

7 Team Details and Responsibilities

The THOTH project is developed by a multidisciplinary team of six computer engineering students with experience in robotics, artificial intelligence, embedded systems, and software development. The team collaborates across several technical domains to design, implement, and test the robotic tour guide system.

System Architecture: Overall system design and integration of hardware and software components.

Autonomous Navigation: Development of robot localization, mapping, and path planning using ROS2 navigation technologies.

Computer Vision: Implementation of visual perception systems for environment understanding and interaction support.

AI Interaction: Development of conversational interfaces using speech recognition, large language models, and multilingual support.

Hardware Integration: Integration of sensors, embedded electronics, computing hardware, and power systems.

Mechanical Design: Design and development of the robot structure and mobility platform.

8 Long-Term Vision

THOTH is designed as a scalable platform that can extend beyond a university prototype and support future applications in cultural and public spaces.

Museum Deployment: The system can be adapted for different museums and exhibitions, allowing autonomous robotic guides to assist visitors and enhance educational experiences.

Heritage Sites: Future versions of the platform could be deployed in outdoor or semi-outdoor historical locations such as the Giza Plateau, providing guided tours for large tourist groups.

Public Service Environments: The core technology can also be adapted for interactive guidance in airports, hospitals, malls, and other large facilities where visitors require navigation assistance.

National Digital Transformation: Projects such as THOTH align with Egypt's Vision 2030 initiatives by promoting the use of artificial intelligence and robotics to digitize and enhance cultural heritage experiences.

THOTH represents a step toward more interactive, technology-driven cultural experiences and smarter visitor engagement systems.